

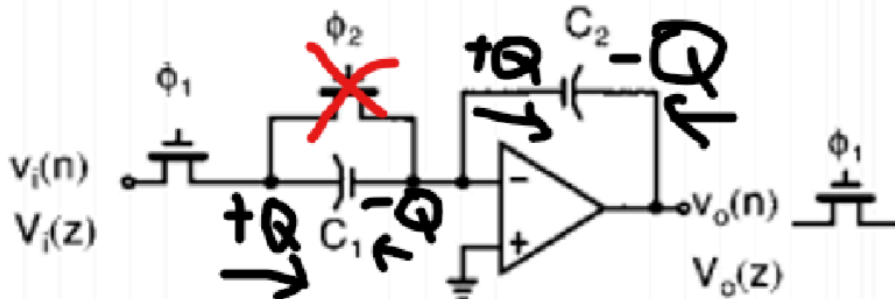
Q1

(a) Neglecting parasitic capacitances, find the discrete time transfer function of the circuit in Fig. 1. Assume that the op amp is ideal

(10 marks)

C1 is discharged during ϕ_2 .

Charge sampled on C1 during ϕ_1 is transferred to C2 via charge conservation at high impedance node.



$$v_o(n) = -(C_1/C_2)v_i(n) + v_o(n-1/2) = -(C_1/C_2)v_i(n) + v_o(n-1)$$

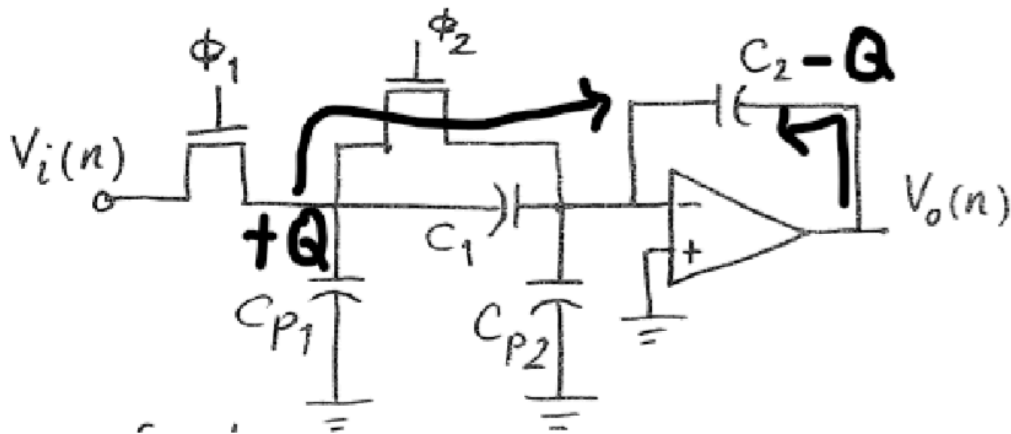
$$V_o(z) = -(C_1/C_2)V_i(z) + V_o(z)z^{-1}$$

$$V_o(z)(1-z^{-1}) = -(C_1/C_2)V_i(z)$$

$$H(z) = V_o(z)/V_i(z) = -(C_1/C_2)/(1-z^{-1})$$

(b) Including parasitic capacitances, find the discrete time transfer function of the circuit in Fig. 1

(10 marks)



C_{p2} has no charge sharing effect since it is virtual grounded. C_{p1} charge is transferred during ϕ_2 .

Consider charge sharing during ϕ_2 ,

Any C_1 charge is cancelled by ϕ_2 shorting switch, so only C_{p1} charge is transferred and this appears in next ϕ_1

$$v_o(n) = -(C_1/C_2)v_i(n) + v_o(n-1/2) - (C_{p1}/C_2)v_i(n-1)$$

$$= -(C_1/C_2)v_i(n) - (C_{p1}/C_2)v_i(n-1) + v_o(n-1)$$

$$v_o(n) - v_o(n-1) = -[(C_1/C_2)v_i(n) + (C_{p1}/C_2)v_i(n-1)]$$

$$V_o(z)(1-z^{-1}) = -[(C_1/C_2)V_i(z) + (C_{p1}/C_2)V_i(z)z^{-1}] = -V_i(z)[(C_1/C_2) + (C_{p1}/C_2)z^{-1}]$$

$$H(z) = V_o(z)/V_i(z) = -[(C_1/C_2) + (C_{p1}/C_2)z^{-1}]/(1-z^{-1})$$

Q2

Following assumption (3), delay of each cell (τ_{inv}) is time required for each output to change by $V_{ref}/2$.

$$v_o(t) = V_{ref}(1 - e^{-\tau_{inv}/RC_L}) = V_{ref}/2 \Rightarrow \tau_{inv} = RC_L \ln 2$$

$$R = \frac{V_{ref}}{I_b}$$

$$T_{osc} = n \tau_{inv} = n RC_L \ln 2 = n \frac{V_{ref}}{I_b} C_L \ln 2 \quad \text{where } n \text{ is the number of delay stages}$$

$$f_{osc} = \frac{I_b}{n V_{ref} C_L \ln 2}$$